

(Please write your Enrolment No. immediately)

Enrolment No: | | | | | | | | | | | | | | | |

MID-TERM EXAMINATION (SET 2)
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
B.Tech — 3rd Semester • September, 2025
Paper Code: CIC-205 • Subject: Discrete Mathematics

Time: 1½ Hours

Maximum Marks: 30

Note: Attempt Q. No. 1 which is compulsory and **any two** more questions from the remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Represent the statement "The alarm sounds if and only if the door is open" using logical connectives. Write its converse, inverse, and contrapositive.	[2]	CO1
1 (b)	Prove that the statement $[p \wedge (p \rightarrow q)] \rightarrow q$ is a tautology using a truth table.	[2]	CO1
1 (c)	Define a permutation and a combination. Also, how many positive integers less than or equal to 200 are divisible by 2 or 7 but not both?	[2]	CO4
1 (d)	Use rules of inference to demonstrate the validity of the following argument. Show how the conclusion is valid: Premises: If it snows, the school closes. The school is not closed. It is cold. Conclusion: It did not snow.	[2]	CO1
1 (e)	Use the Rules of Inference to show that the premises "If it rains, I will take the bus," and "I did not take the bus," and "I went to the store," lead to the conclusion "It did not rain and I went to the store."	[2]	CO1
2 (a)	Find the general solution of the Recurrence Relation: $a_n = 5a_{n-1} - 6a_{n-2} + 2^n$	[5]	CO2
2 (b)	A box contains red, blue, and green marbles. What is the minimum number of marbles we must select to guarantee that at least 9 marbles are of the same color?	[5]	CO2
3 (a)	Let $B = \{5, 10, 15, 20\}$. Find the power set of B, $\mathcal{P}(B)$. Define a Partial Order Relation. Give an example of a relation that is symmetric and transitive but not reflexive.	[5]	CO2
3 (b)	For each of the following functions, determine if it is injective (one-to-one), surjective (onto), and/or bijective. Justify your answer: (I) $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = 3x - 1$ (II) $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = x^3$	[5]	CO2
4 (a)	Simplify the following Boolean expression using a 4-variable Karnaugh map: $F(A, B, C, D) = \sum m(0, 1, 2, 8, 9, 10, 11, 14, 15)$ where $\sum m$ indicates the sum of minterms.	[5]	CO3
4 (b)	In how many ways can a committee of 7 people be formed from a pool of 10 scientists and 8 engineers, if the committee must contain at least 4 engineers?	[5]	CO4